

EDMONTON STONY PLAIN, AB, Canada

WMO: 711270

Lat: 53.5475N

Lon: 114.1083W

Elev: 2514

StdP: 13.41

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-14.6	-27.2	1.3	-19.4	-20.9	1.9	-14.0	22.0	32.0	19.2	31.1	4.3	270	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	81.0	63.0	78.1	61.5	75.3	60.0	65.9	76.9	63.9	74.4	62.1	72.0	5.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.0	91.2	70.6	59.9	84.4	68.8	57.9	78.5	66.7	32.1	77.0	30.5	74.6	29.1	72.2	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	15.6	13.4	DB	-25.3	87.0	6.5	3.7	-29.9	89.6	-33.7	91.8	-37.3	93.9	-42.0	96.7	(4)
			WB	-26.2	69.3	5.7	2.4	-30.3	71.1	-33.6	72.5	-36.8	73.9	-41.0	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	14.6	17.8	25.8	39.7	51.3	58.2	62.9	60.8	52.4	40.5	25.7	15.4
	DBStd	20.65	16.39	14.12	13.56	9.63	8.23	5.69	5.45	5.98	7.93	8.74	12.27	14.67
	HDD50	5343	1099	901	750	326	83	6	0	4	64	311	728	1071
	HDD65	9616	1564	1321	1215	760	427	214	105	155	381	760	1178	1536
	CDD50	1286	0	0	0	17	124	253	401	338	136	17	0	0
	CDD65	82	0	0	0	0	3	11	41	24	3	0	0	0
	CDH74	875	0	0	0	2	61	122	361	251	75	3	0	0
	CDH80	136	0	0	0	0	5	16	63	41	11	0	0	0
Wind	WSAvg	5.9	5.8	5.7	6.0	6.8	6.4	5.7	5.5	5.1	5.6	6.3	5.9	6.2
Precipitation	PrecAvg	17.5	0.8	0.5	0.7	1.2	2.0	3.1	3.5	2.2	1.4	0.9	0.7	0.6
	PrecMax	23.9	2.6	1.3	1.3	2.3	4.4	6.9	7.5	4.7	3.6	3.3	2.2	1.8
	PrecMin	10.3	0.0	0.0	0.1	0.2	0.4	0.7	1.0	1.1	0.2	0.2	0.1	0.0
	PrecStd	3.3	0.6	0.3	0.3	0.6	1.0	1.5	1.5	0.8	0.9	0.7	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	48.9	56.0	71.5	80.3	82.1	85.9	84.4	81.1	70.5	55.6	48.3
		MCWB	37.8	38.2	42.8	51.2	56.5	62.2	65.5	64.4	61.0	53.3	43.7	38.5
	2%	DB	43.3	43.1	49.9	65.4	75.2	77.4	81.1	79.8	75.4	62.5	48.2	42.7
		MCWB	36.1	35.0	39.0	48.0	54.5	59.1	64.7	63.5	58.5	49.4	39.5	35.5
	5%	DB	39.6	39.5	45.9	59.8	71.2	73.8	77.9	76.5	71.0	57.4	43.6	38.9
		MCWB	33.8	32.7	36.9	44.8	53.0	57.8	63.4	61.9	56.5	46.1	36.9	33.1
	10%	DB	35.8	36.0	42.1	54.9	66.6	70.3	74.8	73.2	66.3	53.2	39.9	34.8
		MCWB	31.3	30.6	34.8	42.3	50.6	56.1	61.5	60.3	54.2	43.7	34.5	30.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.7	39.1	43.3	51.9	59.0	64.5	69.3	67.8	62.3	54.1	44.5	39.6
		MCDB	45.5	47.3	54.4	69.8	74.4	75.7	78.9	79.6	77.3	68.9	54.2	47.4
	2%	WB	36.5	35.6	39.9	48.6	56.2	61.5	66.9	65.3	59.5	50.0	40.6	36.0
		MCDB	42.5	42.1	48.7	63.8	71.0	73.2	77.5	76.4	72.8	60.6	47.1	41.5
	5%	WB	34.1	33.1	37.4	45.8	54.0	59.4	64.8	63.3	57.2	47.2	37.8	33.3
		MCDB	39.2	39.0	44.8	58.3	68.1	70.0	75.2	73.7	68.8	55.7	43.1	38.5
	10%	WB	31.4	30.9	35.2	43.0	51.8	57.6	63.0	61.6	55.1	44.7	35.0	30.4
		MCDB	35.8	35.9	41.5	53.0	64.6	67.4	72.8	71.2	65.2	51.8	39.1	34.7
Mean Daily Temperature Range	5% DB	MDBR	14.8	16.1	15.9	18.3	19.9	18.2	18.7	19.3	19.2	16.4	13.1	13.6
		MCDBR	16.7	19.4	20.3	26.7	26.6	23.8	23.2	24.7	27.5	23.9	17.1	15.3
		MCWBR	12.0	13.2	12.0	14.0	12.0	10.5	10.7	11.2	12.9	12.9	10.9	10.9
	5% WB	MCDBR	17.0	19.1	19.5	25.6	24.3	21.1	21.7	23.0	25.8	22.8	16.8	16.1
		MCWBR	12.9	13.4	11.9	14.0	11.4	10.1	11.0	11.6	12.6	12.8	11.3	12.2
Clear-Sky Solar Irradiance	taub	0.249	0.262	0.298	0.327	0.342	0.350	0.353	0.358	0.309	0.281	0.253	0.238	
	taud	2.333	2.370	2.364	2.389	2.390	2.383	2.425	2.361	2.499	2.557	2.462	2.319	
	Ebn at Noon	238	273	283	287	287	285	282	273	277	265	238	219	
	Edh at Noon	17	23	30	33	35	36	34	34	26	20	16	15	
All-Sky Solar Radiation	RadAvg	305	632	1107	1475	1786	1893	1933	1596	1105	634	311	222	
	RadStd	28	49	76	137	130	126	108	109	114	70	31	20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page